JANMITRA

Civic Issue Reporting & Management System

Nagpur Municipal Corporation | 4 Categories | 4 Wards | 4 Departments

Team ResolveIT

Final Tech Stack & Consistency Verification

Verified technology stack and cross-document consistency check

CITIZEN PORTAL	Report Issue | Track Status | Verify Resolution
BACKEND (Django)	REST API | Auth | Business Logic | AI Module
AI VALIDATION	Fake Detection (EXIF+ELA+HuggingFace) | YOLO Relevance Check
DATABASE	PostgreSQL + PostGIS | 4 Categories | 4 Wards | 4 Depts
DEPT. DASHBOARD	PWD | Solid Waste | Water Works | Electrical

Anas Shaikh | Anurag Deshmukh | Om Gadekar | Faizan Rayeen

September 2026 | B.Tech Final Year Project

1. Final Technology Stack -- Complete Reference

INFO This tech stack was critically reviewed against actual project requirements. Only technologies with a clear, necessary role are included. Nothing was added for popularity. Nothing unnecessary was kept.

FRONTEND

Technology	Ver.	Role	Justification
React.js	18	UI framework	One codebase for all 4 portals. Component-based. Industry standard.
Tailwind CSS	3	Styling	Faster than writing CSS. Responsive utilities built-in.
React Router v6	6	Navigation	Client-side routing. ProtectedRoute for role-based access.
React Context API	18	Auth state	Built-in React. Stores token + user + role. No extra library needed.
Axios	-	HTTP client	JWT interceptor + auto-refresh. Simple configuration.
Leaflet	1.9	Maps	Free, no API key. GPS pin for complaints. OpenStreetMap tiles.
React-Leaflet	4	React map wrapper	Makes Leaflet components React-friendly.
Recharts	2	Charts	React chart library. Bar, pie, line charts for dashboard.
Firebase JS SDK	10	Push receive	FCM token request + foreground/background push receive.

BACKEND

Technology	Ver.	Role	Justification
Python	3.11	Language	Same language as AI libs. No separate AI microservice needed.
Django	4.2	Web framework	LTS. ORM, migrations, admin, URL routing all included.
Django REST Frwk	3.15	REST APIs	Serializers, viewsets, RBAC permissions. Standard Django REST.
DRF SimpleJWT	5.3	Authentication	JWT access (60min) + refresh (7d). Permission classes for RBAC.
django-cors-headers	4.4	CORS	Allow React to call Django API in development.
django-environ	0.11	Config	Load env vars from .env safely. One place for all secrets.
Gunicorn	22	Prod. server	Replaces runserver in production. Required on Render.
psycopg2-binary	2.9	DB driver	Python-PostgreSQL driver. Binary for easier install.

DATABASE

Technology	Ver.	Role	Justification
PostgreSQL	15	Primary database	PostGIS support. Best Django ORM compatibility. Free on Render.
PostGIS	3.3	Spatial extension	GPS storage + duplicate detection within 100m in one query.

AI / ML

Technology	Ver.	Role	Justification
Pillow	10.4	EXIF reading	Read camera metadata. Layer 1 fake detection, step 1.
OpenCV (headless)	4.10	ELA analysis	Image compression comparison. Layer 1 step 2. Headless for servers.

NumPy	1.26	Array math	Pixel diff calculation for ELA. Required by OpenCV.
Transformers (HF)	4.44	AI fake detect	Pre-trained model: real vs AI-generated. Layer 1 step 3.
PyTorch	2.3	ML runtime	Required by Transformers. Not used directly.
Ultralytics YOLOv8	8.2	Object detect	Layer 2 relevance check. Fine-tuned on civic issue dataset.

BACKGROUND PROCESSING & NOTIFICATIONS

Technology	Ver.	Role	Justification
Celery	5.4	Async task queue	AI pipeline runs in background. User never waits. Periodic auto-close.
Redis	7	Celery broker	Message queue for Celery. Fastest broker. Free 30MB on Redis Cloud.
Firebase Admin	6.5	Push from backend	Send FCM notifications from Django. Free Firebase Spark plan.

DEPLOYMENT & INFRASTRUCTURE

Technology	Role	Justification
Docker + Compose	Local dev	Celery needs Linux. Docker gives identical env on all Windows machines. One command.
Render Free Tier	Cloud host	Free web service, background worker, static site, PostgreSQL. Zero cost.
Redis Cloud	Hosted Redis	Render has no free Redis. Redis Cloud gives 30MB free. Enough for task queue.
UptimeRobot	Prevent sleep	Pings Render every 10 min. Keeps service awake for demo.
GitHub	Version control	Free private repos. Feature branch + PR workflow for 4 members.

2. Technologies NOT Included (and Why)

Technology	Reason NOT Included
Zustand	React Context API is sufficient for auth state (token+user+role). Unnecessary dependency.
AWS S3 / CloudFront	Images stored on server (Django MEDIA_ROOT). Free. No S3 cost or complexity for prototype.
Stable Diffusion	Dataset augmentation uses existing HuggingFace fake image datasets instead.
Grafana / Power BI	Recharts inside React is simpler and free. Sufficient for 4-metric dashboard.
Kafka	Celery + Redis is the correct choice at our scale. Kafka is enterprise-level.
ElasticSearch	No full-text search needed. Django ORM + PostgreSQL LIKE is sufficient.
Kubernetes	Render manages containers. K8s is far beyond the scope of a college project.
SQLite as fallback	One DB system only (PostgreSQL). No dual-DB strategy. Simplifies setup.
RabbitMQ	Redis is simpler to set up and free. Both work as Celery broker. Redis chosen.
Vue / Angular	React is chosen. No framework mixing. One frontend framework throughout.
Flutter	Responsive React web works on phones. No app install needed. Simpler.

3. Cross-Document Consistency Check

TIP All 9 documents were reviewed for consistency before finalisation. The results are below. All documents use identical terminology, the same 4 categories, the same 4 wards, and the same tech stack.

3.1 Category Consistency

All documents reference exactly these 4 categories and no others:

Category	In PRD?	In SRS?	In Arch?	In Dev?	In Team?	In Dashboard
Roads & Infrastructure	YES	YES	YES	YES	YES	YES
Garbage & Waste Management	YES	YES	YES	YES	YES	YES
Water & Drainage	YES	YES	YES	YES	YES	YES
Street Lighting & Electrical	YES	YES	YES	YES	YES	YES

3.2 Ward Consistency

Item	PRD	SRS	Arch	UI/UX	Worker Doc	Dashboard
Ward count	4	4	4	4	4	4
Ward names	1-4	1-4	1-4	1-4	1-4	1-4
Worker ward FK	YES	YES	YES	YES	YES	YES
Complaint ward FK	YES	YES	YES	YES	YES	YES

3.3 Tech Stack Consistency

Technology	PRD	SRS	Arch	Dev	Stack Doc

React 18	YES	YES	YES	YES	YES
Django 4.2	YES	YES	YES	YES	YES
PostgreSQL+PostGIS	YES	YES	YES	YES	YES
Celery+Redis	YES	YES	YES	YES	YES
YOLOv8	YES	YES	YES	YES	YES
Firebase FCM	YES	YES	YES	YES	YES
Docker+Render	YES	YES	YES	YES	YES
Zustand	NO	NO	NO	NO	NO

Zustand is listed as NO everywhere -- it was removed in favour of React Context API.

3.4 User Roles Consistency

Role	PRD	SRS	Arch	UI/UX	Team	Worker Doc
Citizen	YES	YES	YES	YES	YES	YES
Worker	YES	YES	YES	YES	YES	YES
Dept. Officer	YES	YES	YES	YES	YES	YES
Admin	YES	YES	YES	YES	YES	YES

3.5 Department Consistency

Department	PRD	SRS	Arch	Dashboard	Worker
Public Works Department (PWD)	YES	YES	YES	YES	YES
Solid Waste Management Dept. (SWM)	YES	YES	YES	YES	YES
Public Health Engineering / Water Works	YES	YES	YES	YES	YES
Electrical Department	YES	YES	YES	YES	YES

4. Key System Rules -- Final Reference

WARN These rules must be implemented consistently in code across all modules.

Rule ID	Description
R-01	Exactly 4 civic categories. No additions at runtime.
R-02	Exactly 4 wards. No additions at runtime.
R-03	Exactly 4 departments (one per category). No additions at runtime.
R-04	Category always determines department. No manual dept. selection.
R-05	Citizen selects ward from dropdown. No automatic GPS-to-ward detection.
R-06	Worker must have both department AND ward set. Neither can be null.
R-07	Worker assignment: worker.dept == complaint.dept AND worker.ward == complaint.ward.
R-08	Ward override (R-07 bypass): Admin only. Requires reason text. Logged in status_history.
R-09	Dept. Officer API always filters by request.user.department_id. Cannot be bypassed.

R-10	Worker task API always filters by request.user.department_id AND request.user.ward_id.
R-11	Mark Work Done is blocked until at least 1 proof photo uploaded (frontend + backend).
R-12	Citizen verification: 72-hour auto-close via Celery periodic task if no response.
R-13	All secrets (DB URL, Redis, FCM key) stored in environment variables only.
R-14	All infrastructure uses free/open-source tools. Zero cost.

5. Final Document Index

Doc #	Title	Key Content
1	PRD	Project overview, 4 categories, 4 wards, features, functional requirements
2	SRS	DB schema, use cases, status flow, API list, ER diagram
3	Architecture	3-tier design, AI pipeline, dept+ward routing, deployment, security
4	UI/UX	All screens for all 4 portals, design system, responsive rules
5	Development	Tech stack (reviewed), folder structure, code samples, deployment checklist
6	Team Distribution	Who does what, timeline, Git workflow, coordination points
7	Dept. Dashboard Spec	Dashboard widgets, filters, admin view, API design for dashboard
8	Worker Management	Worker fields, registration, ward+dept assignment rules, override logic
9	Tech Stack + Check	Final tech stack, removed tools, cross-doc consistency verification